OpsArena: Production Cloud-Native Incident Response RL Environment

TECHNICAL SPECIFICATION & VERIFIER DESIGN DOCUMENT | TEAM_087

Team Code: team_087	**Track:** Software & DevOps	**Evaluated Model:** gpt-6-luna (penpencil.guru)	**Pre-Flight Gate:** 8/8 Passed (2.30s)

1. Executive Summary & Problem Formulation

Modern cloud operations require autonomous AI agents to diagnose, triage, and remediate cascading incidents across distributed microservices under strict Service Level Objectives (SLOs: error rate < 0.1%, p99 latency < 100ms) and operational budgets ($150.00). Existing agent benchmarks either evaluate synthetic toy math puzzles or provide unconstrained Docker environments vulnerable to shell escapes, credential leaks, and verifier tampering. OpsArena provides a deterministic, discrete-event simulation of CloudMesh (8 microservices), a dual-container isolated evaluation harness, an 8-task procedurally seeded benchmark suite, and a mathematically gated multi-factor verifier.

2. Dual-Container Architecture & Isolation Boundary

• **Network Segmentation & Least Privilege:** The harness container connects to the external model API and the internal bridge. The sandbox container runs exclusively on an isolated Docker network (internal: true) with zero public egress. It executes as an unprivileged user (UID 1000: sandbox), with cap_drop: ALL, no-new-privileges, a 512MB RAM ceiling, and no Docker socket mounted.
• **Verifier Inaccessibility & Single Source of Truth:** All scoring thresholds and reward logic reside solely inside the harness. The sandbox exposes only /health, /reset, /tool, and /state. The underlying metrics engine serves as the single source of truth for both live model queries and verifier evaluation, guaranteeing 100% agreement without drift.

3. Operational Action Space (8 Schema-Validated Typed Tools)

Tool	Cost	Parameters & Type Bounds	Operational Function & Safeguards
`get_topology`	$1.00	None	Inspects 8 services: pod replicas, versions, dependencies, and cumulative spend.
`query_metrics`	$2.00	service, metric (latency, errors, cpu, mem, qps, rps)	Timeseries metrics with HEALTHY / CRITICAL status. Validates inputs prior to cost deduction.
`query_logs`	$3.00	service, filter (case-insensitive substring)	Structured log query with regex secret sanitization and strict 4,000 char output bounds.
`scale_service`	$2/pod+$3	service, replicas (int: 1 to 20)	Scales pod pool within platform boundaries; rejects non-ints and out-of-range values.
`rollback_release`	$20.00	service, target_version (string)	Rolls back faulty deployments to verified stable container images in release registry.
`update_config`	$5.00	service, key, value (service-scoped typed)	Service-scoped schema: enforces int bounds (pool: 1-100, lock: 0-60k) and booleans. Rejects unlisted keys.
`reroute_traffic`	$10.00	service (order-service), weight (int: 0 to 100)	Shifts canary traffic weights (0% to 100%) to safely isolate failing routes.
`finish`	$0.00	None	Signals mitigation completion, freezes state/budget against further mutation, triggers scoring.

4. Multi-Factor Programmatic Reward Formulation & Mathematical Gating

OpsArena computes a multi-factor continuous reward gated by an absolute incident resolution indicator: **Reward = P × [0.45 × S_SLO + 0.25 × R_reg + 0.15 × B_bud + 0.15 × E_eff]**.
• **Progress Gate (P ∈ [0.0, 1.0]):** P = 0.00 for no-op/idle agents, guaranteeing a score of **0.0000** ≤ **0.05**. Bounded partial credit (P ∈ (0, 0.5]) is awarded for incremental progress. Full resolution (P = 1.00) strictly requires satisfying SLO thresholds (error rate ≤ 0.10% and p99 latency ≤ 100ms). If either SLO is breached, progress is capped at P ≤ 0.40, guaranteeing score cannot exceed 0.40.
• **SLO Recovery (S_SLO ∈ [0, 1]):** Continuous score evaluating post-incident error rate (≤ 0.10%) and p99 latency (≤ 100ms) with dual-layer telemetry agreement.
• **Platform Stability (R_reg ∈ [0, 1]):** Compares non-target microservices against pre-incident baseline health. If an agent downscales, rolls back, drains, or misconfigures an unaffected service, R_reg incurs target-specific penalties (-0.25 to -0.40), preventing uncoordinated cluster mutations.
• **Budget Preservation (B_bud ∈ [0, 1]):** Full marks for spend ≤ $50.00, decaying linearly to 0 at the $150.00 operating budget ceiling.
• **Action Efficiency (E_eff ∈ [0, 1]):** Full marks for surgical resolution within ≤ 8 total model turns, decaying linearly up to 40 turns.

5. Procedurally Seeded 8-Task Incident Benchmark Suite

All 8 task instructions state business symptoms, observable alerts, and SLO constraints without leaking hidden root causes or exact tool commands:

Task ID	Scenario Title	Diff	Simulated Root Cause & Seeded Variations	Target SRE Action	Oracle	No-Op
`task_01`	Redis Eviction Storm	Easy	Mem capacity (1-4GB), surge QPS (2.8k-3.6k) causes DB CPU 96%.	Scale cache-cluster ≥ 2 pods	1.000	0.000
`task_02`	Bad Deployment Rollback	Easy	Faulty release tag throws NullPointerExceptions (16-21% 500s).	Rollback order-service to v2.4.0	1.000	0.000
`task_03`	DB Pool Starvation	Med	Checkout bursts (1.6k-2.2k RPS) exhaust pool (504 timeouts).	Update db_pool_size ≥ 25	1.000	0.000
`task_04`	Slow Third-Party Gateway	Med	External payment partner hangs (6s-9s), causing thread starvation.	Enable circuit_breaker_enabled	1.000	0.000
`task_05`	Cascading Failure	Hard	Flash surge (4.4k-5.2k RPS) overwhelms pods, causing OOM crash loops.	Scale order-service ≥ 6 pods	1.000	0.000
`task_06`	Canary Memory Leak	Med	Severe memory leak on canary split route (5-15% traffic).	Reroute traffic weight to 0%	1.000	0.000
`task_07`	Overprovisioned Drain	Hard	Cluster overscaled (36-42 pods, $540-$630/hr). Safe min enforced.	Right-size cluster to ≤ 18 pods	0.988	0.000
`task_08`	Migration Lock Starvation	Hard	Exclusive table lock blocks writes (22% checkout errors, PID 4k-9k).	Set lock_timeout_ms to 1000	1.000	0.000

6. Empirical Evaluation & Live Autonomous Model Runs

• **Pre-Flight Baseline Gating:** Across all 8 tasks, pre-flight checks completed in **2.30 seconds** (budget: 120s), with 100% pass rate (Oracle Mean: 0.999 ≥ 0.95, No-Op Mean: 0.0000 ≤ 0.05). Invariance verified across multiple seeds (0, 42, 99).
• **Live gpt-6-luna Autonomous Evaluations (penpencil.guru):** Three genuine model runs were executed and archived in test-run-logs/:
1. **run_01_task_02_bad_deployment:** Diagnosed NPE, rolled back to v2.4.0, verified error rate dropped from 18.4% → 0.04% in 4 turns (12.1s, Reward: **0.9955**).
2. **run_02_task_01_cache_pressure:** Investigated cache eviction storm, scaled cache-cluster to 2 pods, verified CPU normalized in 4 turns (11.9s, Reward: **0.9955**).
3. **run_03_task_05_cascading_failure (Negative Control):** Step cap hit during exploratory reads before issuing scaling command in 1 turn (3.3s, Reward: **0.0000**).
• **Sample Trajectories:** Documented in sample-trajectories/trajectory_success_task_02.md and sample-trajectories/trajectory_failed_task_05.md with full SRE post-mortems.

7. Adversarial Verification & Tested Security Controls (28 Automated Tests)

Verification Vector / Attack Scenario	OpsArena Tested Implementation & Defense	Automated Test / Status
Task 7 Undersized Cluster Attack Defeat	Minimum safe capacity enforced per service. Scaling all pods to 1 triggers severe SLO starvation.	`test_task_07_undersized_cluster_attack_defeated [PASS]`
Non-Target Rollback Regression Penalty	Rolling back non-target services (e.g. payment-service in Task 2) incurs strict regression penalty.	`test_target_specific_rollback_penalty_task_02 [PASS]`
Non-Target Downscaling Regression Penalty	Downscaling non-target services in Task 7 incurs regression penalty, blocking cluster starvation.	`test_target_specific_downscaling_penalty_task_07 [PASS]`
Strict Mathematical SLO Gating	error_rate > 0.10% or latency > 100ms caps progress at <= 0.40 and total score at <= 0.40.	`test_strict_slo_scoring_gate_breach_caps_score [PASS]`
Full Config Specification Enforcement	All allowed configuration keys (pools, timeouts, circuits) have strict type and boundary specs.	`test_all_declared_config_key_specs_execute_cleanly [PASS]`
Post-Finish State & Budget Freeze	Calling read tools or finish after finish freezes cost and step counts, while mutations fail.	`test_post_finish_complete_freeze_reads_and_cost [PASS]`
Credential Scrubbing & 4k Bounds	Regex redacts API keys, bearer tokens, and passwords. Output strictly clipped at 4k chars with marker.	`test_secret_scrubbing_and_output_bounds [PASS]`

8. Turnkey Reproduction & Official Run Contract

Single-command execution: docker compose up --build --abort-on-container-exit --exit-code-from harness.
Exit codes: **0** = Success (Oracle passed & model completed), **1** = Model crash / unhandled failure, **2** = Pre-flight baseline gate failure (model never invoked), **3** = Misconfiguration / invalid argument.